

Unsupervised deep surrogate model integrating with finite volume method for solving parametrized groundwater flow equations with varying multi-point sources

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ABSTRACT

Real-time prediction of groundwater dynamics under exploitation is crucial for sustainable groundwater management and decision-making. Traditional numerical models have a solid theoretical foundation and are widely used. However, their complex model development procedures and large computational cost make them difficult for users to operate, especially under the condition with varying pumping well locations and pumping rates. Developing surrogate models is an effective way to provide more convenient and efficient prediction of groundwater flow. In this study, a surrogate model based on deep learning method is developed for parametrized groundwater equations with multiple point sources. The surrogate model can learn the relationships between layouts of point sources (e.g., pumping or injection wells) and groundwater levels in an unsupervised manner. This allows the model to serve as a surrogate for predicting groundwater dynamics under randomly varied point sources with different locations and pumping rates, while without running the computationally intensive numerical model. In our deep surrogate model, the finite volume method (FVM) is integrated into the construction of loss function for guiding the neural networks to train the UNet. The model can be conveniently applied to solve multi-point sources problems with discontinuous diffusion tensor coefficients, while also providing a certain level of theoretical guarantee for the predictions without any precomputed dataset. Four numerical examples with increasing levels of difficulty are used to evaluate our method. Comparisons with data-driven method reveal that: 1) when tackling problems with heterogeneous diffusion coefficients, our model successfully inherits the advantages of FVM, accurately predicting solutions close to numerical references, with a mean L_2 error on the order of $1e^{-2.5}$. 2) when the number of groundwater wells is fixed at four, our method achieves a mean L_2 error on the order of $1e^{-4}$, which is comparable to data-driven methods. 3) As the number of wells increases to eight, our method attains a mean L_2 error of $1e^{-2.5}$, surpassing the data-driven method's $1e^{-2.0}$. 4) As the combinations of well numbers, locations, and pumping rates becomes more complex, our method maintains a performance advantage over data-driven approaches. 5) The data-driven method often produces noisy predictions, while our results are smoother, demonstrating superior generalization ability.

1. Introduction

With the development of society, the importance of groundwater resources has become increasingly prominent (Scanlon et al., 2023). It is an important source of drinking water and irrigation water in many regions. However, these resources are influenced by a variety of complex factors, which pose significant challenges in their protection and sustainable management (Karimidastenaie et al., 2022; Grison et al., 2023).

The groundwater flow equation, derived from the fundamental physical principles of physics including mass balance and Darcy's Law, describes flow through an aquifer (El Mezouary and El Mansouri, 2021). Analytic solutions to these equations are typically obtained under simplifying assumptions, such as the isotropy of the medium, while these assumptions often fail to capture the complexities of real-world groundwater flow (Jaiswal et al., 2022; Lin et al., 2022; Tang et al., 2023). Numerical solutions suitable for large, heterogeneous aquifer with complex

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boundary and source/sink terms, are necessary for practical applications.

Although the numerical modeling for practical groundwater dynamics is computationally costly, rapid progress in computer technology has created new opportunities, for conducting efficient and accurate simulations. Numerous researchers have contributed to the field of effective numerical methods for groundwater flow simulation (Banaei et al., 2021; Mohtashami et al., 2022). Widely-used software tools, such as MODFLOW (Langevin et al., 2017; Langevin et al., 2022; Leaf and Fienen, 2022; Brunner et al., 2010), which employs the finite difference method (FDM), FEMFLOW (Diersch, 2002; 2013; Koskinen et al., 2025), based on the finite element method (FEM), and HydroGeoSphere (Brunner and Simmons, 2012; Talebmorad and Ostad-Ali-Askari, 2022), which utilizes the finite volume method (FVM), have been instrumental in advancing the numerical modeling of groundwater flow. In addition, Qian et al. (2023) adapted the vertex-centered FVM introduced by Zhang et al. (2017) to locally refined meshes for solving the groundwater flows, while Miao et al. (2023) proposed a novel cell-centered FVM capable of handling heterogeneous and anisotropic diffusion problems. Although Traditional numerical algorithms achieve remarkable results in groundwater hydrological modeling, they still encounter difficulties when it comes to solving parametrized partial difference equations (PDEs) (Khoo et al., 2021), which are demanded when addressing practical application problems, such as optimizing the layout of pumping wells and enabling the real-time prediction (Koric and Abueidda, 2023; Jesmani et al., 2015; Ahmed et al., 2014).

Accurately solving parametrized equations is not only a theoretical issue but also a widely recognized challenge, primarily due to the inherent complexity of parameter settings (Pasetto et al., 2013; Vadeboncoeur et al., 2023). Many researchers have tried to use the traditional numerical algorithms to solve the parametrized PDEs. The most common method is the exhaustive algorithm which runs a numerical PDE algorithm directly to solve the equations determined by the new coming parameters (Waldowski et al., 2023). This method presents two major challenges. Firstly, it requires sampling a sufficient number of parameters and then solving the corresponding equations, which consumes a large number of computational resources in solving the parametrized PDEs. Secondly, it often encounters difficulties when dealing with parametrized problems with complex parameter settings, which is a common issue in real-world groundwater modeling. In this work, specifically, the parameters in the equation refer to the various spatial layouts of pumping wells (e.g., numbers and locations) with different pumping rates. The complexity of the parameter settings grows exponentially as the number of wells increasing, a phenomenon known as the “curse of dimensionality” (Hu et al., 2024). Instead of using the conventional numerical algorithms directly, reduced order models are more applicable algorithms for solving parametrized PDEs (Bertrand et al., 2023; McQuarrie et al., 2023; Fresca and Manzoni, 2022). For example, Van Doren et al. (2006) used the proper orthogonal decomposition (POD) method to build a low-order model to speed up optimization of water-flooding in a heterogeneous reservoir with multiple injectors and producers. And, Vermeulen et al. (2004) used a similar approach for fast solving the groundwater flow equation. Basically, the reduced order models rely on the snapshots, which are usually called the labeled dataset in the field of deep learning. The so-called proper orthogonal basis functions are selected or derived from snapshots, and are further used as the interpolation basis functions or Galerkin basis functions to expand a subspace with much smaller dimension than the original function space containing the true solution. However, the suitability of the basis functions selected by the POD algorithm depends entirely on the quality of the dataset. In practice, constructing a sufficiently comprehensive and precise labeled dataset for deriving the basis functions is challenging.

With the development of artificial intelligence (AI) technology, a growing body of research has demonstrated that the artificial intelligence models can achieve higher accuracy with lower computational

costs. It can automatically learn features rather than constructing features or selecting orthogonal bases manually (Haggerty et al., 2023). Deep surrogate models for resolving parametrized PDEs have attracted extensive attention in recent years (Chen et al., 2021a; Fuhg et al., 2023; Pestourie et al., 2023). Many data-driven surrogate models have been developed for solving practical problems. Kim et al. (2020) proposed the convolution neural networks (CNN) to build a deep surrogate model to forecast the production profiles of oil wells. Also, Chu et al. (2020) used a data-driven multi-modal CNN for computing the forwarding problem in the process of optimizing infill well placements. A more sophisticated attention mechanism-based UNet architecture was utilized by Taccari et al. (2022), furnishing promising approximate solutions for 2D steady-state groundwater equations with parametrized boundary conditions and hydraulic conductivities. Zheng et al. (2024) proposed a CNN-based surrogate model, OANW, to surrogate the original forward groundwater model. Admittedly, during the training stage, surrogate models demand substantial computational resources. Nevertheless, once trained, these models can solve equations for new parameters in real-time, significantly reducing computational resources and time (Müller et al., 2021; Luo et al., 2023; Sun et al., 2021). Moreover, these models can proficiently handle the “curse of dimensionality” by learning from a sufficiently comprehensive labeled dataset. When a predefined dataset contains a sufficient amount of labeled data, these data-driven methods offer the greatest flexibility in solving scientific computing problems, as the latent relationships between parameters and solutions are already elucidated in the labeled data (Garzón et al., 2022).

The major challenge for training a deep surrogate model is constructing a labeled dataset, which is often resource intensively and costly, especially in groundwater hydrological problems where the available information is often limited. Unlike purely data-driven methods, physics-informed and theory-guided methods reduce the neural network's dependence on labeled data. Some methods can allow for completely unsupervised training without needing any dataset (Xu and Zhang, 2024; Adombi et al., 2021). For instance, He et al. (2021) proposed a surrogate model for identifying the location and strength of a contaminant source in an aquifer through combining physics-informed and data-driven loss functions, demonstrating that this approach can effectively train a CNN with fewer labeled data while maintaining accuracy. The parametrized PDEs defined in irregular domain can be also solved by the CNN (Gao et al., 2021). Zhang et al. (2022) trained a deep neural network model GW-PINN to solve the parametrized groundwater equation with point sources in an unsupervised manner, relying entirely on the physics-informed loss function.

Although the use of residual loss functions and theory-guided methods can reduce the dependence of neural networks on labeled data, it often leads to the consequence of lower accuracy. Recently, a multi-level neural network was proposed to improve the accuracy of physical-informed methods for simulating the parametrized porous media flow (Song et al., 2025). However, it requires stacking and training multiple networks, resulting in significantly increased computational time and resource consumption during training. Beyond accuracy limitations, another constraint in applying these physics informed methods is their inadequate handling of the heterogeneity and anisotropic, which are common characteristics in groundwater simulations. It is primarily because most loss functions of physics informed methods rely on simple finite difference schemes or the auto-derivative (AD) algorithm (Wang et al., 2022; Xu et al., 2021; Wang et al., 2020; 2021; Chen et al., 2021b; Rong et al., 2022), leading to suboptimal performance on training.

To address the issues mentioned above, this paper aims to integrate conventional numerical methods with deep learning techniques to develop a novel unsupervised surrogate model for solving groundwater equations with parametrized multiple point sources. The key contribution of this work lies in proposing a novel pseudo-labeled data generator compatible with any conventional numerical method. The FVM proposed is used as the backbone conventional numerical method for

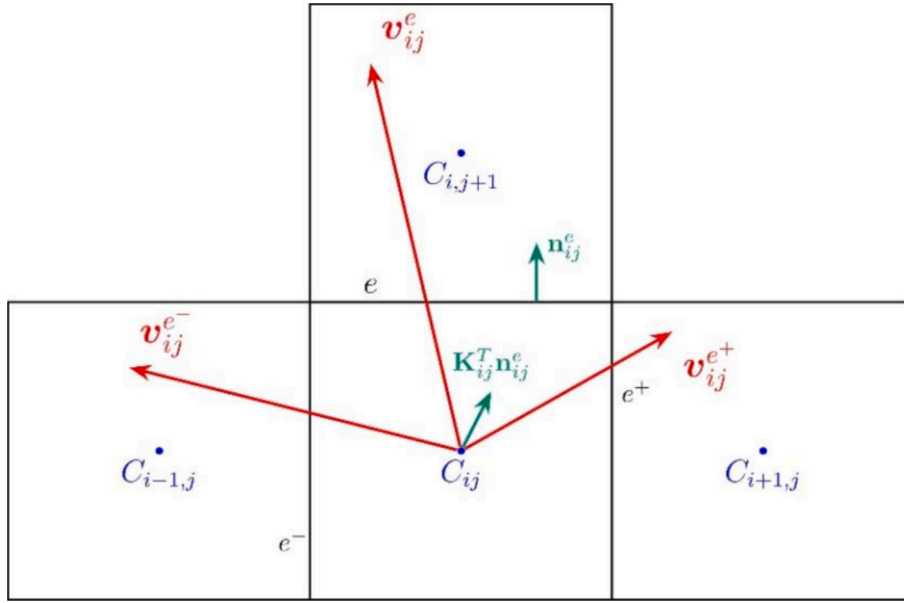


Fig. 1. The diagram of the one side flux of the cell element C_{ij} .

illustrating our deep surrogate model. The CNN architecture employed in this study is consisting of an encoder for feature extraction, a decoder for feature reconstruction and approximated solution prediction. Our model is designed to handle complex scenarios, including anisotropic and heterogeneous porous media. Multiple cases are implemented to test the model accuracy and efficiency. The remainder of this paper is organized as follows. Section 2 introduces the groundwater flow equation and the finite volume method which will be used to construct the loss function. Section 3 provides a detailed explanation of our deep surrogate model, including the UNet architecture, the pseudo-labeled data generator, and the training strategy. Section 4 introduces a series of parametrized problems with increasing complexity. Section 5 presents our numerical results in detail, followed by Section 6 with conclusion.

2. Groundwater flow equation and the finite volume method

This section outlines the mathematical formulation of the groundwater flow problem and details the finite volume method used.

2.1. Steady-state groundwater equation

In this study, we focus on the steady-state groundwater flow with multiple point sources, and the governing equations are expressed as follows,

$$-\nabla \cdot \mathbf{K} \nabla h = f(x), x \in \Omega \quad (1)$$

$$h(x) = g_D(x), x \in \Gamma_D \quad (2)$$

$$-\mathbf{K} \nabla h(x) \cdot \mathbf{n}(x) = g_N(x), x \in \Gamma_N, \quad (3)$$

where h is the hydraulic head ([L]), and g_D and g_N represent the specified Dirichlet and Neumann boundary values, respectively. It is important to note that the forms of $\mathbf{K}$ in Eq. (1) are determined by the type of aquifer. For a confined aquifer, it is expressed as $\mathbf{K} = \mathbf{K}_s M$ with M representing the aquifer thickness ([L]). For an unconfined aquifer, it is given by $\mathbf{K} = \mathbf{K}_s h$. Additionally, $\mathbf{K}_s$ denotes the saturated hydraulic conductivity tensor, which is given by,

$$\mathbf{K}_s = \begin{pmatrix} K_{xx} & K_{xy} \\ K_{yx} & K_{yy} \end{pmatrix}, \quad (4)$$

where K_{xx} and K_{yy} are the saturated hydraulic conductivities ([LT^{-1}]) in the x and y directions, respectively, while K_{xx} and K_{yy} present the cross-conductivity terms ([LT^{-1}]).

The term f on the right-hand side of equation Eq. (1) represents the source/sink term, expressed as,

$$f = P - E - \sum_{i=1}^{N_w} Q_i \delta(x - \omega_i), \quad (5)$$

where $P = P(x)$ denotes the recharge to groundwater from precipitation infiltration ([LT^{-1}]); $E = E(x)$ represents groundwater evapotranspiration ([LT^{-1}]); Q_i is the pumping capacity of a well located at ω_i ; N_w denotes the total number of pumping wells, and $\delta(\bullet)$ is the Dirac delta function, defined as,

$$\delta(x - \omega) = \begin{cases} 0, x \neq \omega \\ \infty, x = \omega \end{cases} \text{ and } \int_{\mathbb{R}^2} \delta(x - \omega) dx = 1. \quad (6)$$

The Dirac delta function Eq. (6) is defined as zero everywhere except at a single point, representing a strong singularity. This characteristic makes it challenging to represent or process directly on a discrete grid. To address this, the Gaussian function is commonly employed as an approximation,

$$\delta(x - \omega) \simeq \tilde{\delta}(x - \omega) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\|x - \omega\|^2}{2\sigma^2}\right), \quad (7)$$

where σ is the standard deviation. A smaller σ produces a more sharply peaked function, thereby providing a closer approximation to the Dirac delta function. Generally, σ is chosen to have a linear relationship with the grid spacing.

2.2. Finite volume scheme

In this section, we introduce a finite volume scheme for solving groundwater flow problems, specifically the interpolation-free cell-centered finite volume method (IFCD-FVM) proposed by Miao et al. (2023). This method is particularly suitable for diffusion equations involving discontinuous diffusion tensors, anisotropy, and heterogeneity.

Although the scheme can be applied to arbitrary polygonal meshes, we focus on rectangular meshes within the rectangular solution domain

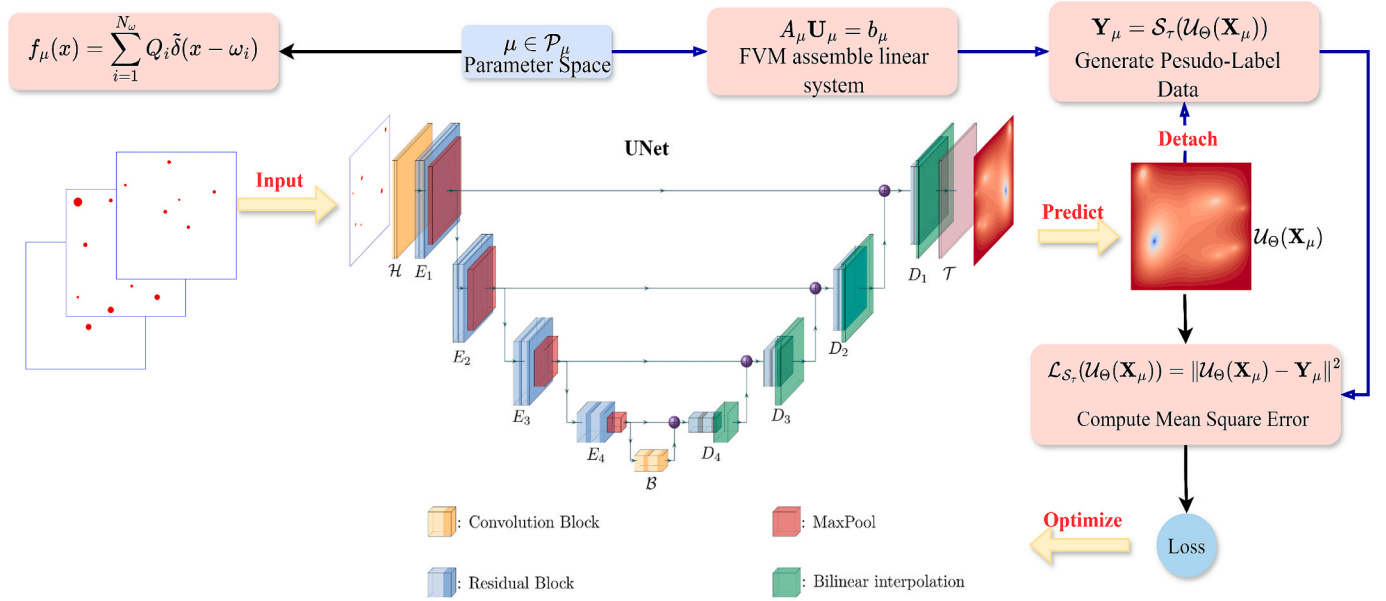


Fig. 2. The flowchart of our deep surrogate model.

$\Omega = [a, b] \times [c, d]$ to align with the structure of convolutional neural networks, in which a, b, c and d represent left, right, lower and upper sides. Next, we briefly derive the IFCD-FVM scheme on a rectangular mesh. To do this, we discretize the domain Ω and obtain a cell-centered rectangular mesh $\mathcal{M} = \{C_{ij}\}_{ij=1}^{N_x, N_y}$ (Note: C_{ij} represents ij -th cell; N_x and N_y are the number of intervals over x and y axes.), where the cell centers are denoted as $\{x_{ij}\}_{ij=1}^{N_x, N_y}$ and x_{ij} is the average of all ij -th cell (four) vertices. As shown in Fig. 1, we use e or $e^{\pm}$ denote the edges, where e^+ and e^- are the adjacent edges of e in the cell C_{ij} , and $\mathbf{n}_{ij}^e$ represents the unit vector normal to e outward to C_{ij} . The $\mathbf{v}_{ij}^e$ is the so-called construction vector and is introduced in the following context. $\mathbf{K}_{ij} = \mathbf{K}(x_{ij})$ is the coefficient tensor at x_{ij} .

2.2.1. Interior edge

The edge flux of the cell element is illustrated in Fig. 1. In the context of an interior edge shared by two cells with C_{ij} and $C_{i,j+1}$ serving as illustrative example, the one-sided flux flowing through e from C_{ij} to $C_{i,j+1}$ is denoted by $\tilde{F}_{ij}^e$ which is defined as,

$$\tilde{F}_{ij}^e = \int_e -(\mathbf{K}_{ij} \nabla h) \cdot \mathbf{n}_{ij}^e ds, \quad (8)$$

and the one-sided flux associated with the cell $C_{i,j+1}$, flowing from $C_{i,j+1}$ to C_{ij} is defined as,

$$\tilde{F}_{ij+1}^e = \int_e -(\mathbf{K}_{ij+1} \nabla h) \cdot \mathbf{n}_{ij+1}^e ds. \quad (9)$$

Thus, we can define the two-sided flux on this edge as,

$$F_{ij}^e = \frac{1}{2} (\tilde{F}_{ij}^e - \tilde{F}_{ij+1}^e), \quad (10)$$

$$F_{ij+1}^e = \frac{1}{2} (\tilde{F}_{ij+1}^e - \tilde{F}_{ij}^e), \quad (11)$$

These two fluxes clearly satisfy the local conservation property $F_{ij}^e + F_{ij+1}^e = 0$.

The difference between various finite volume schemes lies in the construction of the one-sided fluxes $\tilde{F}_{ij}^e$. The IFCD-FVM introduces the so-called construction vectors $\mathbf{v}_{ij}^e$ on the edges e, e^+ and e^- to approximate

the one-sided flux $\tilde{F}_{ij}^e$. Among these, the construction vector on the edge e is defined as,

$$\mathbf{v}_{ij}^e = (x_{i,j+1} - x_{ij}) \cdot \mathbf{n}_{ij}^e \left(1 + \frac{\mathbf{K}_{ij} - \mathbf{K}_{ij+1}}{2\mathbf{n}_{ij}^e \cdot \mathbf{K}_{ij} \mathbf{n}_{ij}^e} \right) \mathbf{n}_{ij}^e. \quad (12)$$

If there is no ambiguity, we simplify the notation $\mathbf{v}_{ij}^e$ as $\mathbf{v}$. The other two construction vectors $\mathbf{v}^+$ and $\mathbf{v}^-$ are defined similarly by replacing the e into $e^{\pm}$. As proven by Miao et al. (2023), the introduction of the construction vector $\mathbf{v}$ ensures the following relationship,

$$h(x_{ij}) - h(x_{i,j+1}) = \nabla h|_{ij} \cdot \mathbf{v}, \quad (13)$$

where $h(x_{ij})$ is the unknown function value at coordinate x_{ij} and $\nabla h|_{ij}$ is the gradient of h at x_{ij} . Using these three construction vectors, two distinct combinations of the co-normal vector $\mathbf{K}_{ij} \mathbf{n}_{ij}^e$ can be obtained,

$$|e| \mathbf{K}_{ij} \mathbf{n}_{ij}^e = \alpha_{ij}^- \mathbf{v}^- + \beta_{ij}^- \mathbf{v}, \quad (14)$$

$$|e| \mathbf{K}_{ij} \mathbf{n}_{ij}^e = \alpha_{ij}^+ \mathbf{v}^+ + \beta_{ij}^+ \mathbf{v}, \quad (15)$$

where $|e|$ is the length of edge e and the combination coefficients are given by,

$$\alpha_{ij}^{\pm} = \frac{|e| \mathbf{v} \cdot \mathbf{K}_{ij} \mathbf{n}_{ij}^e}{\mathbf{v} \cdot \mathbf{v}^{\pm}}, \beta_{ij}^{\pm} = \frac{|e| \mathbf{v}^{\pm} \cdot \mathbf{K}_{ij} \mathbf{n}_{ij}^e}{\mathbf{v}^{\pm} \cdot \mathbf{v}}. \quad (16)$$

To balance the contribution of these construction vectors, we average the two combinations mentioned above, yielding,

$$|e| \mathbf{K}_{ij} \mathbf{n}_{ij}^e = \frac{1}{2} [\alpha_{ij}^- \mathbf{v}^- + (\beta_{ij}^- + \beta_{ij}^+) \mathbf{v} + \alpha_{ij}^+ \mathbf{v}^+]. \quad (17)$$

Therefore, using Eq. ((13)–(15)), we can approximate the one-sided flux Eq. ((8)–(9)) by,

$$\tilde{F}_{ij}^e = \frac{1}{2} [\alpha_{ij}^+ (h_{i+1,j} - h_{ij}) + \alpha_{ij}^- (h_{i-1,j} - h_{ij}) + (\beta_{ij}^- + \beta_{ij}^+) (h_{i,j+1} - h_{ij})], \quad (18)$$

where $h_{ij} = h(x_{ij})$.

2.2.2. Boundary edge

For the edges on boundaries, the definition of construction vector is extended by,

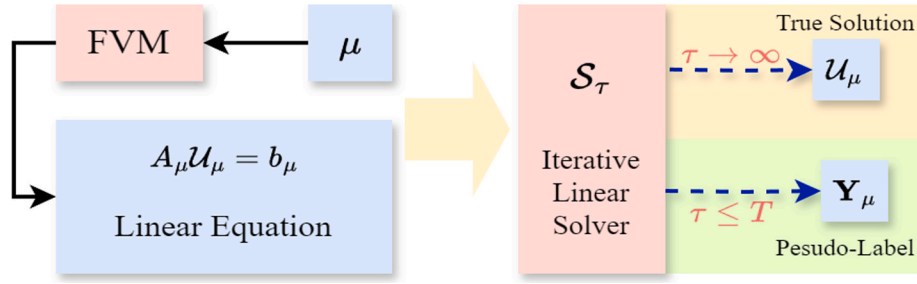


Fig. 3. The flow chart of the Pseudo-Label generator.

$$\mathbf{v} = \mathbf{x}_e - \mathbf{x}_{ij}, \quad \forall e \in \mathcal{M} \cap \partial\Omega, \quad (19)$$

where $\mathbf{x}_e$ is the midpoint of edge e . The edge fluxes of boundary edges are simplified approximated by the one-sided flux, $F_{ij}^e = \tilde{F}_{ij}^e$. Finally, for every element of the mesh we have the following equation,

$$\sum_{e \in \partial C_{ij} \cap \Gamma_N} F_{ij}^e = f_{ij} - \int_{\partial C_{ij} \cap \Gamma_N} g_N ds. \quad (20)$$

where $f_{ij} = f(\mathbf{x}_{ij})$.

By expressing the two-sides flow in Eq. (20) as a linear combination of the unknown function values h_{ij} through Eqs. ((10)–(11)) and Eq. (18), it become to a system of linear equations. The novel pseudo labeled data generator is going to be constructed based on linear systems assembled by the IFCD scheme.

3. Deep surrogate model

In this section, we provide a detailed introduction to the structure of the proposed deep surrogate model, including the network architecture, loss function, and training strategy. The proposed neural network architecture is based on the UNet and designed for learning the map from pumping well layouts to the corresponding solutions of the groundwater equation, as shown in Fig. 2. The model input, denoted by f_μ , are images generating by embedding the layouts of wells, and the outputs, represented by $\mathcal{U}_\Theta(f_\mu)$, are predicted by UNet for approximating the numerical results provided by FVM. The UNet architecture is the backbone network of the deep learning model, which is optimized by minimizing the MSE between the pseudo-label data, denoted by $\mathbf{Y}_\mu$, and predictions. It should be noted that the proposed model aims to be compatible for solving parametrized groundwater flow equations with varying multiple point sources. In specific, the model's input f_μ represents the well layout generated by up to N_w pumping wells with different pumping capacities and locations, forming the parameter space $\mathcal{P}_\mu$,

$$\mathcal{P}_\mu = \left\{ f_\mu(\mathbf{x}) = \sum_{i=1}^{N_w} Q_i \delta(\mathbf{x} - \omega_i) | n_w \in \{1, \dots, N_w\}, Q_i \in [Q_{\min}, Q_{\max}], \omega_i \in \Omega \right\}, \quad (21)$$

where μ denotes the parameter vector consisting of the number of wells N_w , the pumping capacities $\{Q_i\}_{i=1}^{N_w}$ and positions $\{\omega_i\}_{i=1}^{N_w}$.

The training procedure consists of the following key steps: 1) sampling parameters; 2) embedding the parameters into image representations; 3) feeding the images into the UNet and making predictions; 4) generating pseudo-labels, and 5) computing the mean squared error (MSE) and optimizing the UNet via stochastic gradient descent. A detailed description of each step and associated module will be provided in the subsequent sections.

3.1. Pseudo label generator

Unlike the residual loss functions commonly used in the recently popular PINN methods, we propose a novel concept of a pseudo-label generator based on the conventional numerical method. This design is motivated by two key considerations: 1) numerical experiments show that using the residual loss function alone may lead to training failure, preventing the surrogate model effectively learning useful information; 2) The finite volume method is inherently well-suited for handling discontinuities, heterogeneous media, and anisotropic diffusion problems; however, it is rarely employed and difficult to integrate into neural network training frameworks. By introducing the pseudo-label generator, we seamlessly integrate the surrogate model with FVM, significantly improving training efficiency and achieving satisfactory predictive performance. The core idea of the pseudo-label generator is to use the iterative linear solver to correct the predicted solution from the neural network. It is designed to generate pseudo-label data by incompletely solving the linear system of the discretized PDE.

Given a parameter vector μ and the corresponding linear system $\{\mathbf{A}_\mu; \mathbf{b}_\mu\}$ assembled by the FVM. These equations are usually solved by iterative linear solvers, such as Jacobi, Gauss-Seidel and GMRES (Generalized minimal residual methods) algorithm. We use $\mathcal{S}_\tau$ to denote an iterative linear algebra solver with maximal iteration steps τ . As illustrated in Fig. 3, the stiffness matrix assembled using the FVM or other conventional numerical schemes is typically large in size and sparse in structure. Consequently, convergent iterative algorithms are commonly employed to solve the resulting linear system, corresponding to the limit $\tau \rightarrow \infty$. To obtain high-accuracy numerical solutions, supplying a well-informed initial guess is often more computationally efficient than simply increasing the number of iterations in the linear solver. This consideration is especially important for second-order iterative algorithms, where a properly chosen initial condition can substantially reduce computational cost while enhancing solution accuracy. In this context, a well-optimized UNet can serve as an efficient initializer by providing reasonably accurate predictions, enabling the iterative solver to produce pseudo-labels that closely approximate the ground truth using only a few iterations.

Therefore, the prediction $\mathcal{U}_\Theta(\mathbf{X}_\mu)$ of the neural network is used as the initial guess for the iterative solver $\mathcal{S}_\tau$, and the solution obtained after a small number of iterations T is considered as the pseudo-label. With the pseudo labeled data, we can train the neural network to learn the correct direction. The proposed loss function is defined as,

$$\mathcal{L}_{\mathcal{S}_\tau}(\Theta) = \text{MSE}(\mathcal{U}_\Theta(\mathbf{X}_\mu); \mathcal{S}_\tau(\mathcal{U}_\Theta(\mathbf{X}_\mu); \mathbf{A}_\mu, \mathbf{b}_\mu)), \quad (22)$$

where MSE denotes the standard mean squared error function. For simplicity, in this paper, we use the Jacobi iterative method as the solver for the linear system, which can be described as,

$$\mathbf{x}^{(\tau+1)} = \mathbf{D}^{-1}(\mathbf{b} - (\mathbf{L} + \mathbf{U})\mathbf{x}^{(\tau)}), \tau = 1, 2, \dots, T, \quad (23)$$

where $\mathbf{D}$, $\mathbf{L}$ and $\mathbf{U}$ represent the diagonal, lower triangular, and upper triangular parts of $\mathbf{A}$, respectively. The other iterative solvers, such as

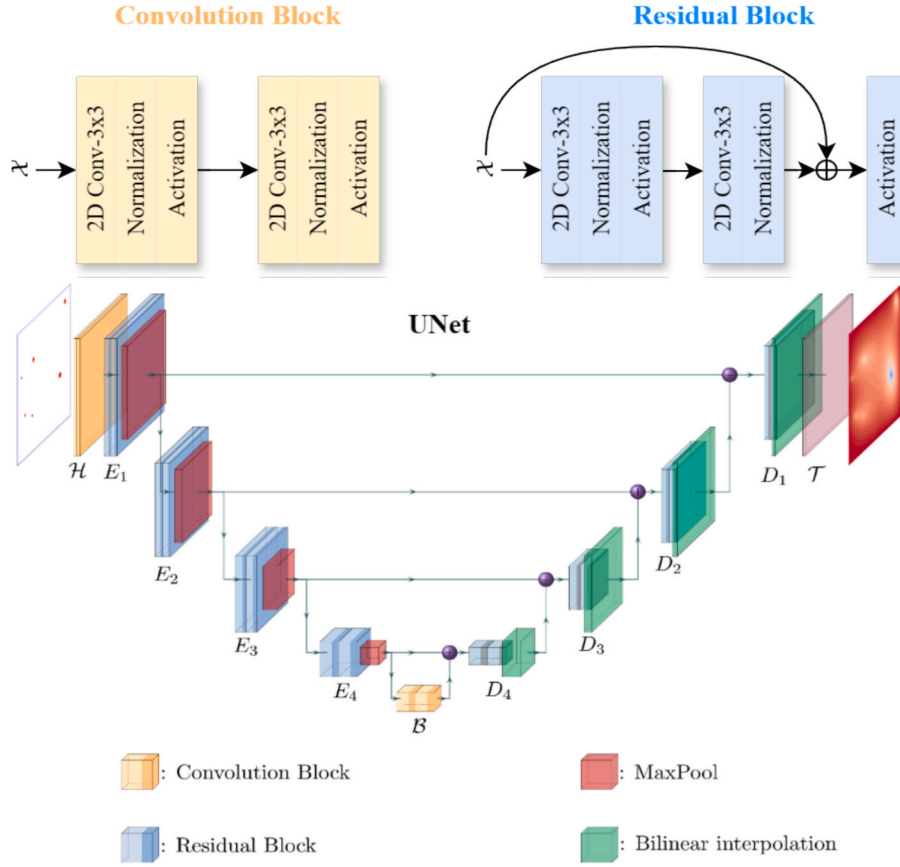


Fig. 4. The proposed UNet's architecture. The architecture includes the convolution block and residual block. We propose to use the Layer Normalization and Tanh activation function as the basic setting.

Gauss-Seidel, gradient-descent, can also be used in our method. Selecting different iterative solvers to generate pseudo label data may affect the performance of the deep surrogate model. Here, we choose to use the Jacobi iterative method because it meets two important conditions: 1) The Jacobi iterative method can solve the linear equations assembled by the FVM. 2) The computation process of Jacobi is simple, involving only matrix–vector multiplication, which can be easily accelerated using GPUs. Therefore, if computational complexity is not a concern, GMRES, which has a faster convergence rate but is more complex, might be more suitable for the other parameterized problems and numerical schemes.

Unlike the PINNs approach, the proposed loss function uses an iterative linear algebra solver to produce pseudo-labels rather than applying PDE or discrete PDE operators directly onto the neural network. Firstly, this avoids the instability caused by automatic differentiation or discrete PDEs. Secondly, it completely decouples traditional numerical algorithms from training deep surrogate models. As a result, our method greatly reduces the computational complexity of PINNs during back-propagation and achieves a more stable training process.

3.2. Network architecture of the surrogate model

The UNet architecture was initially applied to the semantic segmentation of images by [Ronneberger et al. \(2015\)](#), yielding excellent results. The original UNet can also be slightly modified to address scientific computing problem, which has been successfully used in [Lauzon \(2024\)](#), [Hua et al. \(2023\)](#). In this study, the UNet architecture is adopted and the details are shown in [Fig. 4](#). As illustrated in [Fig. 4](#), the network follows the encoder and decoder structure, which is denoted as UNet ($N_1, \dots, N_L; P$), where L is the number of levels, $\{N_l\}_{l=1}^L$ represents the number of residual blocks, and P is the number of feature channels output by the input head $\mathcal{H}$. In the other words, the more levels L or planes P , the

deeper or wider UNet is, which leading to more complex UNet with more trainable parameters.

In specific to the proposed network in this study, the encoder comprises $L = 4$ submodules which are labeled as $\{E_l\}_{l=1}^4$, and the decoder has an equivalent number of submodules denoted by $\{D_l\}_{l=1}^4$. The encoder and decoder constitute a hierarchical structure, which is an essential characteristic of the UNet. Each encoder submodule E_l receives its input from either the output of the preceding encoder module E_{l-1} or, at the first level, from the input header $\mathcal{H}$. The bottleneck module $\mathcal{B}$ lying between the encoder and decoder processes features from the encoder and passes them to the decoder. Subsequently, each decoder submodule D_l produces an output that is directed either to the subsequent decoder module D_{l-1} or, at the final level, to the output tail $\mathcal{T}$. Each encoder-decoder submodule pair, E_l and D_l , is connected by a residual connection, instead of the copy-and-concatenate approach in the original UNet.

In our network architecture, the residual and convolution blocks serve as the fundamental building blocks, upon which all the submodules are constructed. As illustrated in the top of [Fig. 4](#), the convolution block consists of two two-dimensional (2D) convolution operators with a kernel size of 3×3 followed by normalization and nonlinear activation functions, which facilitate robust training and nonlinearity. Similarly, the residual block is constructed by incorporating a residual connection into the convolution block. The input head $\mathcal{H}$ is composed of two stacked convolution blocks. The features produced by $\mathcal{H}$ have P channels and the same spatial size as its input $\mathbf{X}_\mu$. The encoder submodule E_l , built by stacking N_l residual blocks, doubles the number of feature channels and halves the spatial size using a max-pooling operator with a kernel size of 2×2 and a stride of 2. The decoder submodule D_l , also consisting of N_l stacked residual blocks, halves the number of

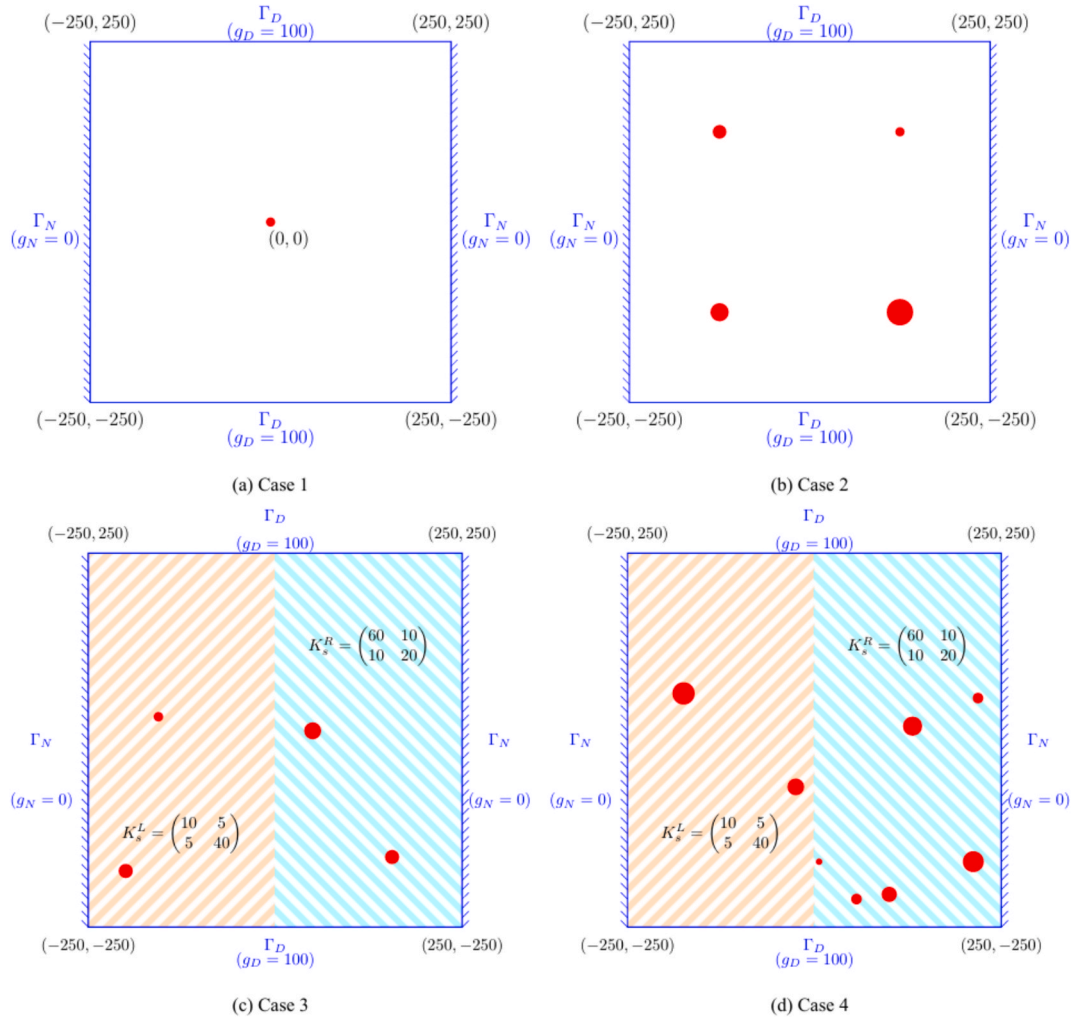


Fig. 5. The problem settings for the four test cases, which progressively increase in complexity, are as follows: the first two involve isotropic media, and the last two involve heterogeneous anisotropic media. The red nodes represent the well location and the varying node size represents the pumping capacity. The unit of length: m; the unit of K_s : m/d. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

feature channels and doubles the spatial resolution using a 2D bilinear interpolation operator. Finally, the output tail $\mathcal{T}$ contains two convolution blocks, without activation function in the end.

3.3. Training strategy

This section introduces the complete training process of the deep surrogate model, as shown in Fig. 2. At the beginning of each training step, we first sample a batch of parameters μ from the parameter space $\mathcal{P}_\mu$; then, we use the FVM to discretize the corresponding PDE to generate the stiffness matrix A_μ and the right-hand vector b_μ , while feeding the embedded image $\mathbf{X}_\mu$ of μ into the UNet to obtain the predicted solution $\mathcal{W}_\Theta(\mathbf{X}_\mu)$; next, we use the iterative linear solver $\mathcal{S}_\tau$ to correct the predicted solution, thereby generating pseudo-label data; finally, we compute the mean squared error (MSE) between the predicted results and the pseudo-labels, and train the UNet using the Adam

optimizer to update the network parameters Θ . It is worth mentioning that during training, we must detach the gradient of the pseudo-labels, otherwise the training will fail. A detailed description of the training process can be found in Algorithm 1.

It is worth noting that as the max iteration $\tau \rightarrow \infty$, the solution obtained by the iterative solver can be considered as the reference solution. At this situation, our method can be considered as the data-driven method which is not favorable. Instead, τ is chosen to be very small, and the iterative solver only acts as a corrector. It guides the neural network to learn in the right direction but not completely solves the problems. Clearly, how to control the number of τ need to be carefully considered in the training process. To help train the neural network but not use too much computation power, a detailed discussion of τ will be illustrated in subsequent sections.

Algorithm 1: Training strategy of the deep surrogate model.

Data: The parameter Space $\mathcal{P}_\mu$, the mesh $\mathcal{M}$.

Input: Number of epochs $E_{\max}$; Number of maximal iterations steps τ .

for epoch = 1, 2, ..., $E_{\max}$ **do**

for $\mu \in \mathcal{P}_\mu$ **do**

 Embedding μ into image format by $\mathbf{X}_\mu = f_\mu(x), x \in \mathcal{M}$;

 Feed $\mathbf{X}_\mu$ into UNet to make predictions $\mathcal{U}_\Theta(\mathbf{X}_\mu)$;

 Assemble stiffness matrix A_μ and right-hand vector b_μ by FVM;

 Generate Pseudo-Label Data $\mathbf{Y}_\mu = \mathcal{S}_t(\mathcal{U}_\Theta(\mathbf{X}_\mu); A_\mu, b_\mu)$;

 Compute the MSE $\|\mathcal{U}_\Theta(\mathbf{X}_\mu) - \mathbf{Y}_\mu\|^2$

 Update Θ by Adam optimizer;

Result: The parameter Θ of the UNet.

4. Problem settings

To validate the proposed method, we selected four parametrized problems with progressively increasing complexity, spanning from simple to intricate scenarios. These problems carry distinct practical significance and serve to thoroughly test the method's adaptability and performance across diverse conditions, and also to provide deeper insights into its strengths and limitations when tackling problems of varying complexity. All problems incorporate Neumann boundary conditions, which are among the most commonly encountered in real-world applications, particularly in fields such as fluid dynamics and heat transfer. The specific configurations and parameter settings for each problem are detailed below.

4.1. Case 1: a single fixed-position well in isotropic media with variable pumping capacity

The first test case serves as a baseline to evaluate the proposed method in a relatively simple scenario, as shown in Fig. 5(a). This case considers a single pumping well located at the center of the domain Ω , specifically at $\omega = (0, 0)$. The source term is expressed as,

$$f_\mu(x) = Q\delta(x - \omega), \quad (24)$$

where Q is the pumping capacity, which is allowed to vary between 100 m^3/d and 1500 m^3/d . This defines the parameter space as $\mathcal{P}_\mu = \{Q | Q \in [100, 1500]\}$, allowing us to investigate how the method handles variations in pumping capacity.

The porous media is isotropic, characterized by a saturated hydraulic conductivity tensor of $\mathbf{K}_s = 33.3\mathbf{I}$ m/d, where $\mathbf{I}$ is the identity matrix, and the aquifer thickness is set as $M = 3$ m. This setup isolates the effect of the variable pumping capacity on the flow behavior, ensuring that variations in the medium properties do not confound the results. The boundary conditions are set as that the top and bottom boundaries are Dirichlet-type boundaries with a constant head value $g_D = 100$ m, and the left and right boundaries are water barriers modeled as Neumann conditions (no flow across these boundaries $g_N = 0$).

This case is ideal for testing the fundamental accuracy and stability of the proposed method in a controlled environment with a single well and isotropic media. By analyzing the results, we can validate the method's baseline performance, which will serve as a comparison for more complex cases.

4.2. Case 2: four fixed-position pumping wells in isotropic media, each with a variable pumping capacity

Building on the first case, this scenario introduces greater complexity by considering four fixed-position pumping wells located at specific coordinates within the domain Ω . The wells are placed at $\omega_1 = (-125, -125)$ m, $\omega_2 = (125, -125)$ m, $\omega_3 = (-125, 125)$ m, $\omega_4 = (125, 125)$ m, as depicted in Fig. 5(b). The source terms are now represented as,

$$f_\mu(x) = \sum_{i=1}^4 Q_i \delta(x - \omega_i), \quad (25)$$

where Q_i is the pumping capacity of the well located at $\{\omega_i\}_{i=1}^4$ and each pumping capacity is allowed to vary independently within the range 500 m^3/d to 1500 m^3/d . This leads to the following parameter space for the pumping capacities,

$$\mathcal{P}_\mu = \{Q_i | Q_i \in [500, 1500], i = 1, 2, 3, 4\} \quad (26)$$

The medium remains isotropic, and the boundary conditions are the same as in Case 1. This case adds the complexity of multiple wells and their interactions for testing the proposed method's ability to model multi-well systems. By independently varying the pumping capacities of each well, this case provides insights into the interactions between multiple wells and their impact on the groundwater flow, which is crucial for real-world applications for managing water extraction from multiple sources.

4.3. Case 3: pumping wells in heterogeneous porous media, with the locations and pumping capacities variable

This case introduces significant complexity by considering a heterogeneous porous medium, where both the well locations and pumping capacities are variable under the fix well numbers. As shown in Fig. 5(c), the medium consists of anisotropic hydraulic conductivities. On the left side of Ω , the saturated hydraulic conductivity tensor is $\mathbf{K}_s^L = \begin{pmatrix} 10 & 5 \\ 5 & 30 \end{pmatrix}$ m/d, and on the right side, it is $\mathbf{K}_s^R = \begin{pmatrix} 60 & 10 \\ 10 & 20 \end{pmatrix}$ m/d. The aquifer thickness and boundary conditions are consistent with previous cases. The source term is defined as,

$$f_\mu(x) = \sum_{i=1}^4 Q_i \delta(x - \omega_i), \quad (27)$$

where the locations $\{\omega_i\}_{i=1}^4$ are randomly generated within Ω , and the pumping capacities $\{Q_i\}_{i=1}^4$ are chosen randomly from the range [100, 1500] m^3/d . This results in the parameter space as,

$$\mathcal{P}_\mu = \{(Q_i, \omega_i) | Q_i \in [100, 1500], \omega_i \in \Omega, i = 1, 2, 3, 4\} \quad (28)$$

To reflect realistic constraints in well placement, we impose a minimum distance of 25 m between any two wells or between a well and the boundary. This case tests the method's ability to handle heterogeneous media with anisotropic hydraulic conductivities, variable well locations and pumping capacities.

4.4. Case 4: pumping wells in heterogeneous porous media, with the number, locations, and pumping capacities all varying

This case represents the most complex scenario, as shown in Fig. 5(d). It builds on the same heterogeneous media as Case 3, maintaining the anisotropic hydraulic conductivity tensors and the aquifer thickness. However, the complexity is significantly increased by allowing the number, locations, and pumping capacities all free. The source term in this case is expressed as,

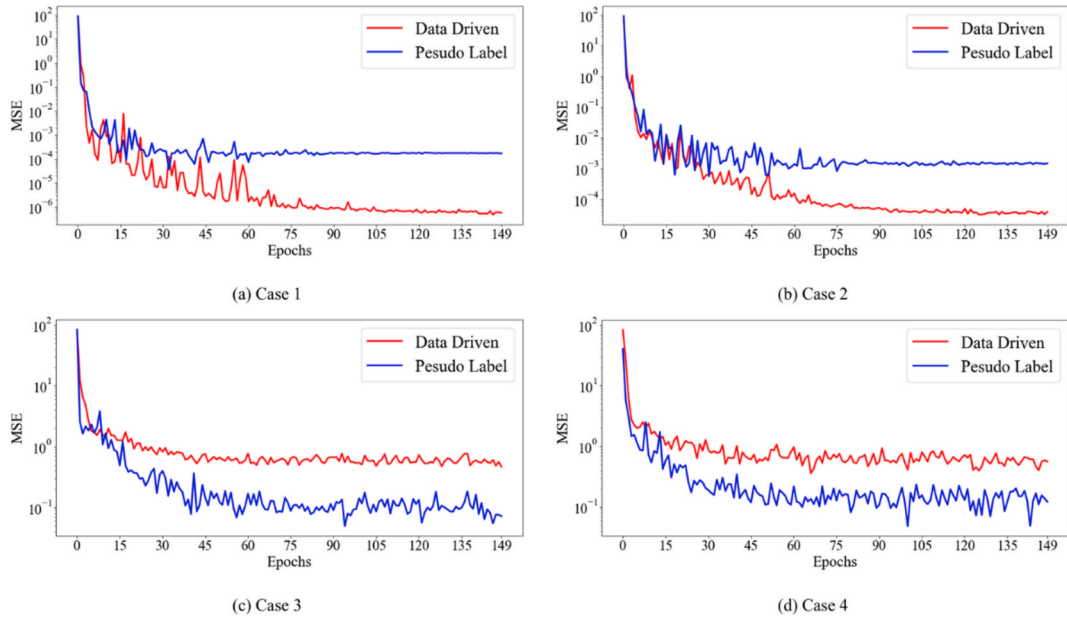


Fig. 6. MSE errors during validation phase of the 4 cases from the data-driven and the propose methods.

$$f_{\mu}(x) = \sum_{i=1}^{N_{\omega}} Q_i \delta(x - \omega_i), \quad (29)$$

where N_{ω} can vary from 1 to 8 (the max number is set 8 in this case considering the study area), with the pumping capacities Q_i ranging from 100 m³/d to 1500 m³/d, and the locations ω_i are randomly generated within Ω while satisfying the same minimum distance constraints as Case 3. This leads to the parameter space,

$$\mathcal{P}_{\mu} = \{(Q_i, \omega_i) | Q_i \in [100, 1500], \omega_i \in \Omega, i = 1, \dots, N_{\omega}, N_{\omega} \in \{1, \dots, 8\}\}. \quad (30)$$

This case introduces a high degree of variability, making it an essential scenario for testing the proposed method's ability to handle a large number of wells with varying numbers, locations and capacities. The flexibility allows for the exploration of more realistic groundwater systems, such as large-scale aquifer management or scenarios with multiple competing water extraction sources. Additionally, the randomness in well placement and pumping capacities simulates real-world uncertainties and optimization-driven decisions.

The boundary conditions remain consistent with Case 1, ensuring comparability across test cases. This scenario serves as a comprehensive test of the method's scalability and robustness, evaluating its performance in complex real-world groundwater management scenarios.

5. Numerical experiments

In this section, we present a comparative analysis using four numerical test cases to validate the effectiveness of the proposed deep surrogate model against data-driven methods. Unless otherwise specified, the computational domain for all cases is a rectangular region $\Omega = [-250, 250] \times [-250, 250]$ m. Details regarding the iteration steps τ of the pseudo-label generator, the key parameters of the network architecture, and the size of the training dataset denoted by N_{train} are all provided as part of ablation studies. Based on these findings and considering the trade-off between computational efficiency and accuracy, the following configurations were selected. The iteration steps were set to $\tau = 5$; the network architecture was chosen as a UNet model with 5 layers and 8 planes, marked as UNet ([2, 2, 2, 2, 2], 8), and the training dataset consisted of $N_{train} = 10,000$ instances. Additionally, 1,000 instances were used for validation, and another 5,000 instances

were used to evaluate the test performance of the trained deep surrogate model. The model is trained using the Adam optimizer with an exponentially decaying learning rate strategy over $E_{max} = 150$ epochs, with an initial learning rate of $1e^{-2}$ and a batch size of 5. For convenience, we use the numerical solution provided by the Finite Volume Method (FVM) as the reference solution, denoted as $\mathbf{h}^*$. The predictions from the deep surrogate model trained in a data-driven paradigm are denoted as $\mathbf{h}^{data}$, and the results from our method are denoted as $\mathbf{h}^{pse}$. In order to ensure the accuracy of the numerical results, the space resolution of the uniform mesh $\mathcal{M}$ is set to be $N_x = N_y = 256$, and all numerical experiments shown following share the same space resolution.

To evaluate the performance of the data-driven method and our approach, the solutions of FVM are treated as benchmark, and three criteria are employed. The first one is the mean squared error (MSE), which is used during the validation for logging the network's performance in training, as,

$$\text{MSE}(\mathbf{h}) = \frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} |h_{ij} - h_{ij}^*|^2, \quad (31)$$

where $\mathbf{h} = \{h_{ij}\}_{i,j=1}^{N_x, N_y}$ means the output of the data-driven method or our approach. The second criterion is the relative L_2 error, which is used to assess the cumulative error across the domain, as,

$$L_2(\mathbf{h}) = \left(\frac{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} |h_{ij} - h_{ij}^*|^2}{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} |h_{ij}^*|^2} \right)^{\frac{1}{2}}. \quad (32)$$

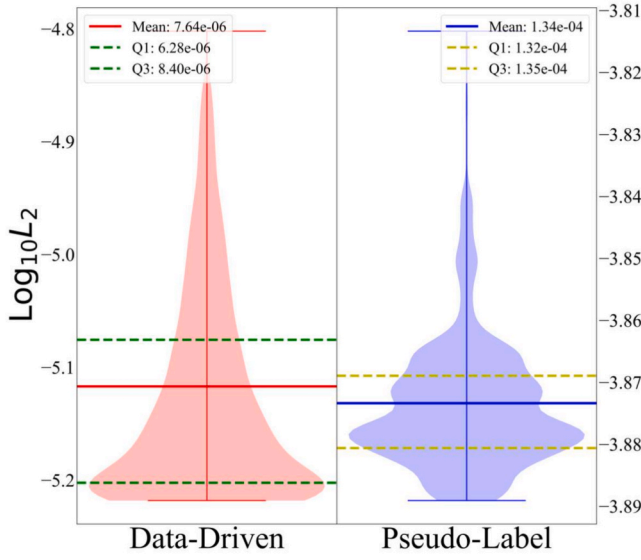
And the third criterion is the absolute error map (AEM),

$$\text{AEM}(\mathbf{h}) = |\mathbf{h} - \mathbf{h}^*|, \quad (33)$$

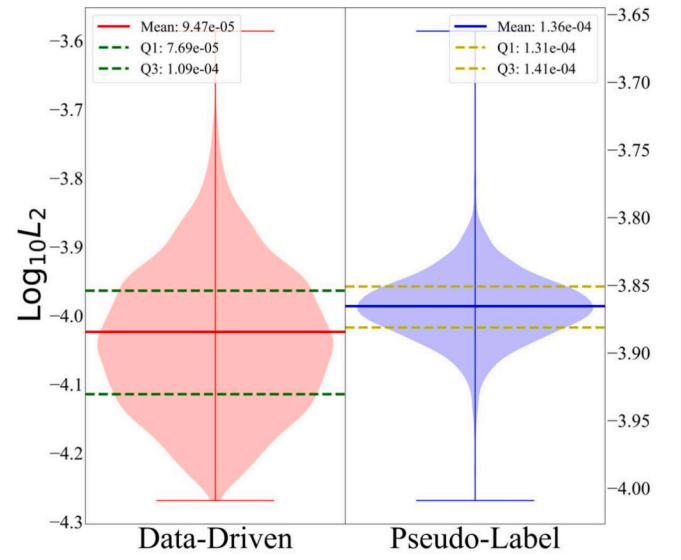
where $|\mathbf{h} - \mathbf{h}^*|$ represents the element-wise absolute value of the difference between $\mathbf{h}^*$ and $\mathbf{h}$. This map provides a visual representation of the magnitude of the errors across the entire domain.

5.1. Validation curve comparisons of the data-driven and the propose methods

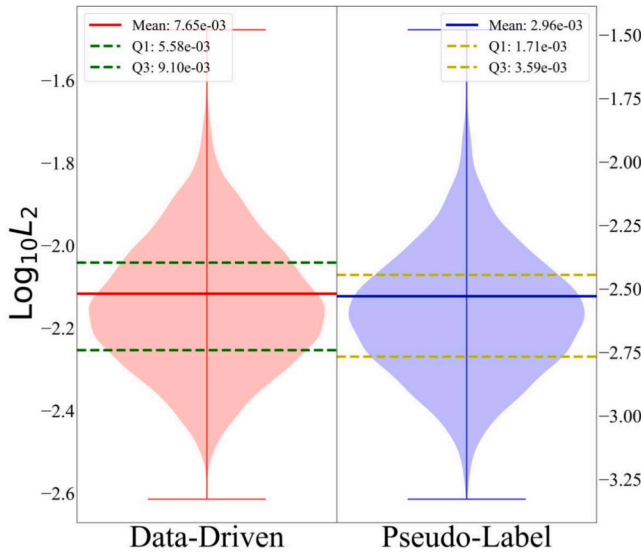
To objectively evaluate the performance of the data-driven and the proposed methods, we plot validation curves for four cases in Fig. 6,



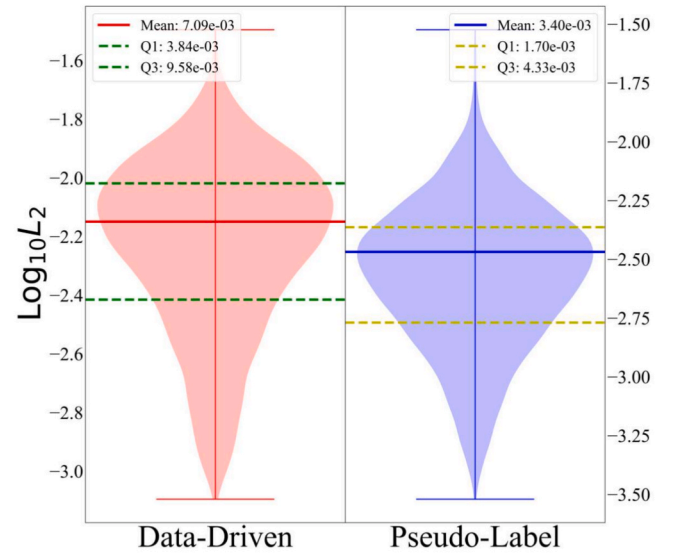
(a) Case 1



(b) Case 2



(c) Case 3



(d) Case 4

Fig. 7. Statistical results for the four cases, including mean, median values and the 25-th and 75-th percentiles.

demonstrating their performance on the validation set evolves with the training process. It is evident that as the complexity of the scenarios increases, our method gradually demonstrates advantages over the data-driven approach. Specifically, in Case 1, due to the relatively simple scenario, the data-driven model achieves an extremely low validation error, with the MSE error ultimately dropping to the order of 10^{-6} . Although our method performs slightly less effectively, it still maintains a relatively low error level around 10^{-4} , as shown in Fig. 6(a).

This result indicates that the data-driven method is better suited for handling simpler problems. However, as the complexity of the scenarios increases, the advantages of our method become more apparent. From Fig. 6(b), it can be observed that in Case 2, our method performs comparably to the data-driven method, with both achieving stable MSE

errors on the order of 10^{-4} during validation. Moreover, in the later stages of training, our method even slightly outperforms the data-driven approach. For Cases 3 and 4, where the complexity significantly increases, our method demonstrates clear superiority, with the final MSE error nearly an order of magnitude lower than that of the data-driven method, as illustrated in Fig. 6(c) and (d). These experimental results fully confirm that our method exhibits stronger robustness and superiority in handling complex problems, as it is more effective at capturing and modeling high-complexity features.

5.2. Statistical results of the data-driven and the proposed methods

To more comprehensively evaluate model performance, we present

Table 1

Statistical results of relative L_2 errors for the ablation study of the UNet architecture.

Levels planes	3		4		5	
	Mean	Median	Mean	Median	Mean	Median
4	2.93e-3	2.54e-3	1.51e-3	1.21e-3	1.17e-3	9.26e-4
8	1.65e-3	1.31e-3	1.43e-3	1.08e-3	1.24e-3	8.45e-4
12	1.70e-3	1.26e-3	1.88e-3	1.50e-3	1.13e-3	8.49e-4
24	1.58e-3	1.11e-3	1.39e-3	1.05e-3	1.28e-3	8.66e-4

violin plots of the relative L_2 error distributions for both methods on the test set in Fig. 7, providing a more complete representation of the relevant statistical information. It is important to note that a violin plot is a data visualization tool that combines a box plot with kernel density estimation. It simultaneously reflects the central tendency, distribution range, and probability density of the data, offering a richer representation of information compared to traditional box plots.

As shown in Fig. 7(a) and (b), although the data-driven method achieves a lower mean error on the test set for Case 1 and Case 2, its error distribution between the 25-th and 75-th percentiles is wider, indicating limited generalization capability across the entire dataset. In contrast, while our method exhibits a slightly higher mean error than the data-driven approach, its error distribution is more concentrated, with the 25-th and 75-th percentiles closer to the mean. This result demonstrates that our method exhibits stronger generalization capabilities in the first two cases. Fig. 7(c) and (d) display the statistical results for the third and fourth test cases, which are significantly more complex than the first two. Notably, the shapes of the violin plots for the data-driven method and our method are quite similar, suggesting comparable generalization performance between the two approaches. However, the violin plots for our method are consistently lower overall than those of the data-driven method, further confirming the superior accuracy of our method in more complex scenarios.

5.3. Ablation studies

In this section, we conduct ablation studies focusing on three key factors that may significantly impact the performance of the deep surrogate model: the UNet network architecture, the number of iterations of the pseudo-label generator, and the scale of the dataset.

Traditionally, it is believed that a UNet architecture with a greater number of planes P and deeper layers L , which includes more trainable parameters, exhibits enhanced learning capabilities. However, this belief is primarily derived from research on data-driven methods, and its applicability to the physics-informed paradigm proposed in this paper remains to be verified. Intuitively, a higher number of iterations τ for the pseudo-label generator leads to more accurate pseudo-label data but also incurs higher computational costs. How to balance these two aspects requires in-depth analysis. Additionally, while a rich dataset typically enhances the performance of data-driven methods, the cost of data acquisition often becomes a training bottleneck. In contrast, our proposed method eliminates the need for labeled data, as training data can be easily obtained through sampling. Nevertheless, considering the impact of overfitting, the influence of dataset scale on model performance still necessitates further investigation. We will address these questions one by one in the following sections.

To ensure a fair evaluation of model performance, all models are trained according to the settings described at the beginning of Section 5, and we report the mean and median of their relative L_2 errors on the test dataset. Clearly, models with smaller mean and median errors on the test set demonstrate superior performance.

5.3.1. Ablation study of the UNet's architecture

The ablation study on the UNet's architecture primarily focuses on the impact of the number of levels L and the number of planes P . We use

Table 2

Statistical results of relative L_2 errors for the ablation study of the Normalization methods and activations functions in surrogate model.

Normalization Activation	Batch		Layer	
	Mean	Median	Mean	Median
ReLU	1.05e-2	9.82e-2	3.21e-3	2.96e-3
Tanh	6.42e-3	5.27e-3	1.24e-3	1.00e-2

Table 3

Statistical results of relative L_2 errors for the ablation study of the iteration number in pseudo-label generator.

τ	Mean	Median	τ	Mean	Median
1	1.65e-3	1.25e-3	20	2.22e-3	1.86e-3
5	1.24e-3	1.00e-2	40	1.60e-3	1.23e-3
10	1.46e-3	1.03e-3	80	1.45e-3	1.06e-3

the fourth case introduced in Section 4 to investigate the influence of the architecture on the performance of the deep surrogate model. In the experiment, τ is set to 5, with 10,000 samples in the training set, 1,000 samples in the validation set, and 5,000 samples in the test set.

The corresponding numerical results are presented in Table 1, from which three main observations can be drawn: 1) With a fixed number of planes P , as the number of levels L increases, the model performs better on both metrics; 2) With a fixed number of levels L , as the number of planes P increases, the model also performs better on both metrics; 3) By selecting a moderately sized network architecture, our surrogate model can be trained effectively and accurately. These three observations not only demonstrate the robustness of our method but also indicate that a UNet architecture with more planes and levels is more likely to predict accurate solutions. To balance computational cost and prediction accuracy, we recommend using $L = 5$ and $P = 8$, which is the model used in Section 5.

5.3.2. Ablation study of the normalization method and activation function

The normalization methods and activation functions are crucial for deep surrogate models. They provide effective assistance for the stable training of surrogate models and nonlinearity. They are both common components in deep learning models, being widely used in a variety of deep learning applications. Here, in order to more fully discuss the impact of the components of UNet on its performance, we conducted ablation experiments on normalization methods and activation functions.

We tested the performance of UNet ([2,2,2,2,2]; 8) with BatchNorm, LayerNorm methods and ReLU, Tanh activations on the fourth problem. The numerical results with $\tau = 5$ are shown in Table 2. Based on the numerical results, it is obviously to see that the Tanh activation function and Layer normalization method is the best combination for our deep surrogate model. Also, our method maintains acceptable results on different combinations.

5.3.3. Ablation study of the maximal iteration steps

The number of iterations τ used in the pseudo-label generator is another key hyperparameter that may significantly impact the performance of the deep surrogate model. We also use the fourth case introduced in Section 4 to investigate the influence of the maximum iteration steps τ on the performance of the deep surrogate model. In the experiment, the network architecture adopts UNet ([2, 2, 2, 2, 2]; 8) recommended in the previous section, with 10,000 samples in the training set, 1,000 samples in the validation set, and 5,000 samples in the test set.

The relevant numerical results are shown in Table 3, from which it can be seen that the model's performance with different τ . It can be clearly seen from Table 3 that when $\tau = 5$, the deep surrogate model performs the best on test dataset, with both the mean and median at the

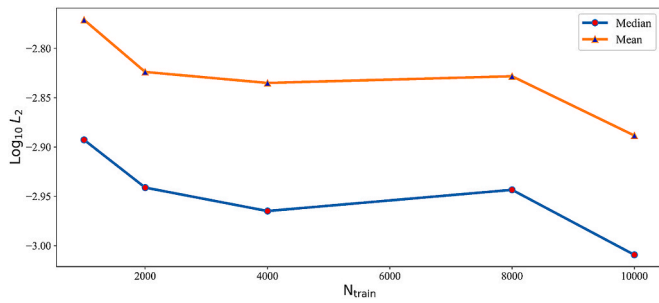


Fig. 8. Statistical results of relative L_2 errors for the ablation study on dataset scale, denoted by N_{train} .

lowest levels. When we increase τ , the performance of the deep surrogate model does not improve significantly but fluctuates within a certain range, especially when comparing the results of $\tau = 1$ and $\tau = 80$. Through our analysis, the possible reasons for this result are as follows: 1) The convergence speed of the Jacobian iteration method is slow, and merely increasing τ by an order of magnitude cannot bring about significant changes, which also indicates that our method is essentially different from the data-driven method. 2) In order to utilize the GPU more efficiently, we chose to use single-precision FLOAT instead of double-precision for training, which limits the accuracy of the surrogate model globally.

However, the robust performance of the deep surrogate model with different τ provides a significant advantage for training, as we can use a very small τ for incomplete iterations when generating pseudo-labels during training, thereby reducing computational costs. This to some extent validates our previous assertion that the Jacobi iteration does not provide true solutions, but rather serves as a guiding mechanism to steer the surrogate model toward better training directions. Thus, $\tau = 5$ is used

in this study.

5.3.4. Ablation study of the scale of the dataset

In data-driven methods, the scale of the labeled dataset plays a pivotal role in model training, while its high acquisition cost also presents a significant bottleneck. Reducing or even eliminating reliance on labeled datasets has become a prominent topic in deep learning research. Fortunately, our approach follows an unsupervised paradigm, enabling the effortless generation of training datasets at various scales by sampling the parameters of partial differential equations (PDEs), without the need for any labeled data. This ablation study aims to investigate the influence of the number of PDE parameters used for training on model performance.

We continue our analysis using the fourth case. To balance computational efficiency and accuracy, based on the outcomes of the previous two ablation experiments, the network architecture employs UNet ([2, 2, 2, 2]; 8), with the iteration number τ set to 5. We select training sets of varying scales, with sample sizes of 10,000, 2,000, 4,000, 8,000, and 10,000, while the validation and test sets are fixed at 1,000 and 5,000 samples, respectively.

The results of this ablation experiment, as illustrated in Fig. 8, reveal the following: 1) The model's performance remains relatively stable across datasets of different scales, and even with only 1,000 samples for training, the model can still produce accurate solutions; 2) With the increase of the training set size, the performance of the deep surrogate model can be improved accordingly. Although our method can generate datasets of arbitrary scale, considering training costs, we recommend using a dataset with a sample size of 10,000.

5.4. Prediction results using the proposed deep surrogate model

To more fully demonstrate the performance of our deep surrogate

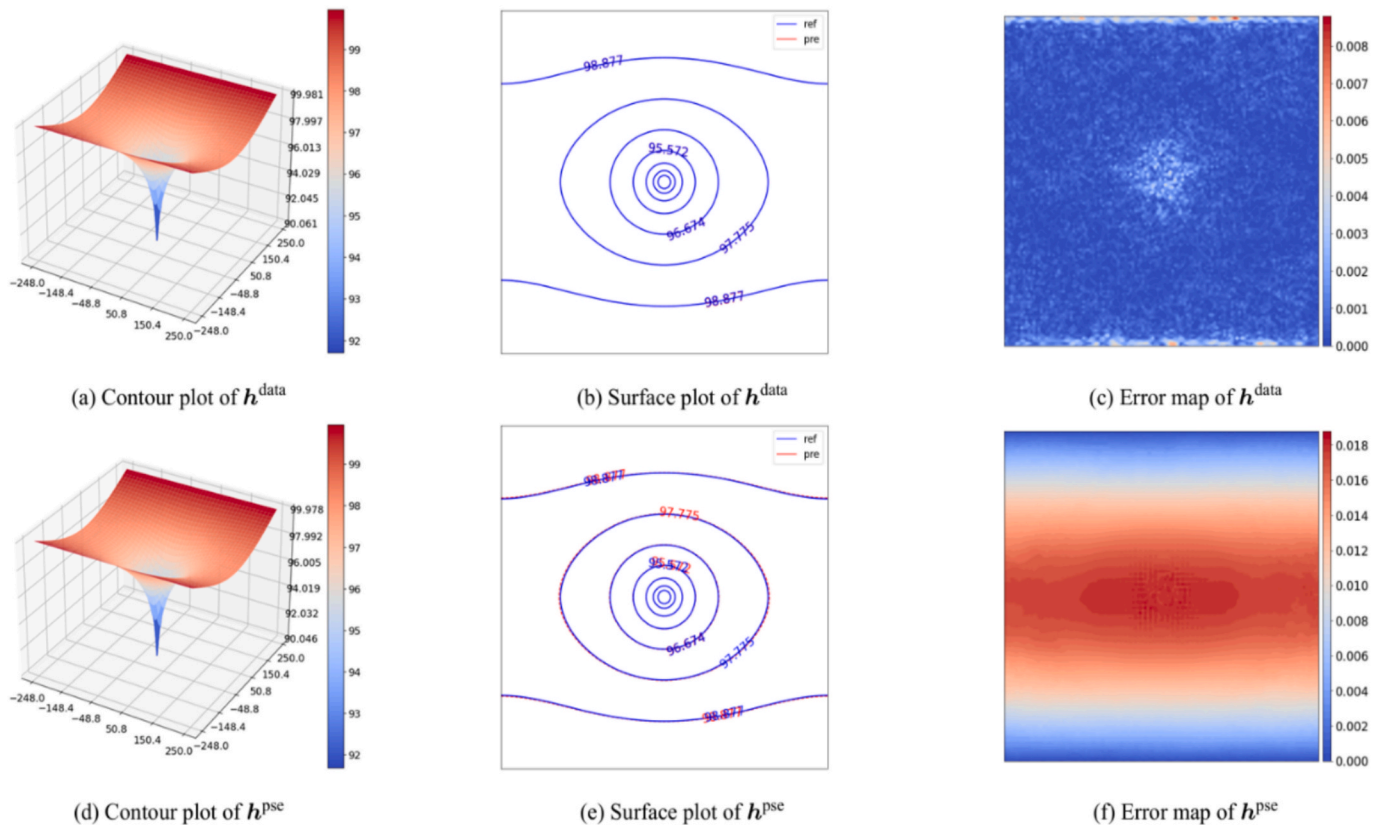


Fig. 9. The comparison of predicted results of Case 1 involving 1 pumping well positioned at the center of the computational domain with pumping capacity of $1320.8 \text{ m}^3/\text{d}$.

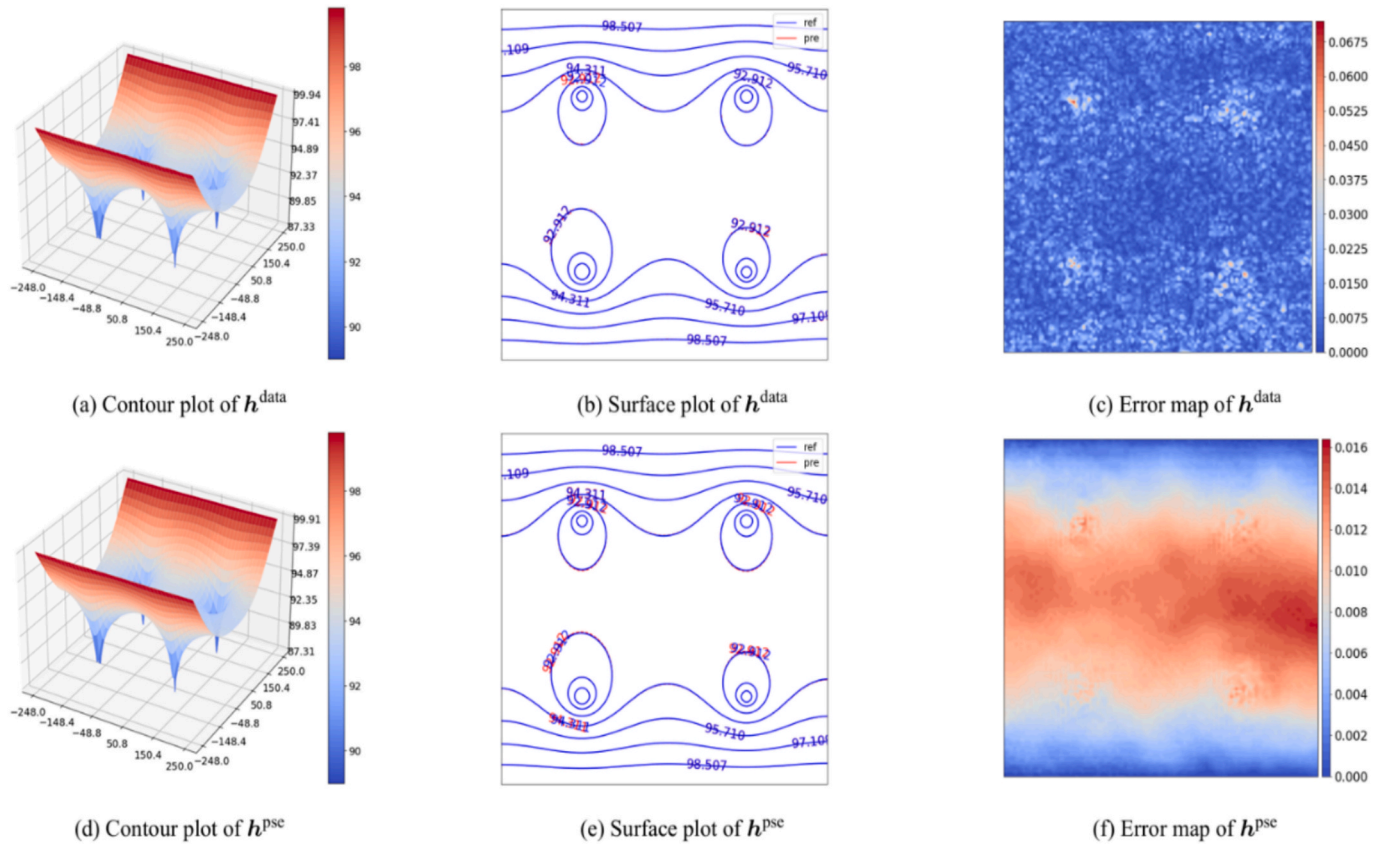


Fig. 10. The comparison of predicted results for Case 2 involving 4 pumping wells positioned at fixed locations within the computational domain. Their pumping capacities are 1445.2, 1281.7, 1350.0, and 1280.4 m³/d, respectively, arranged in a clockwise order starting from the bottom-left corner.

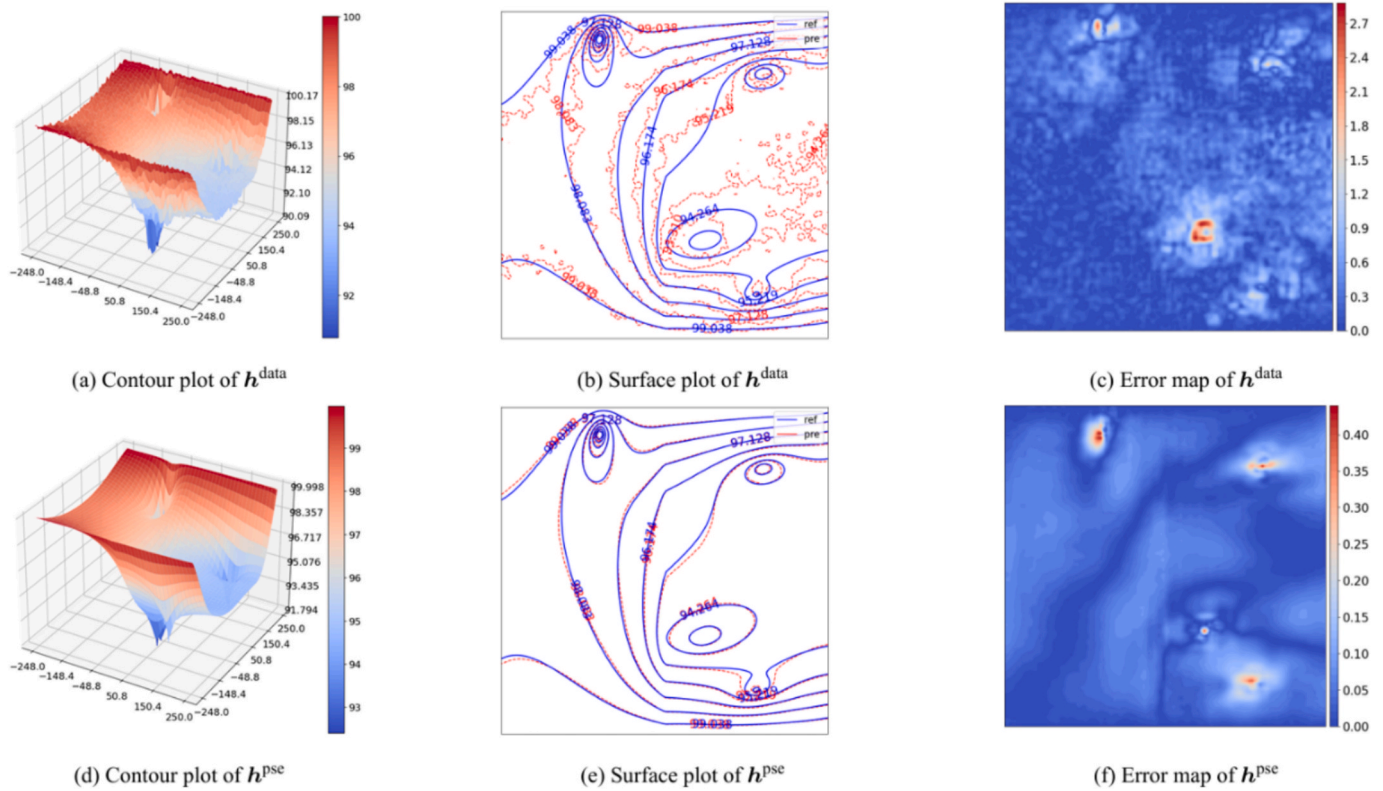


Fig. 11. The comparison of predicted results for Case 3 involving 4 pumping wells randomly distributed within the computational domain. Their pumping capacities are 670.8, 609.9, 824.1, and 802.5 m³/d, respectively.

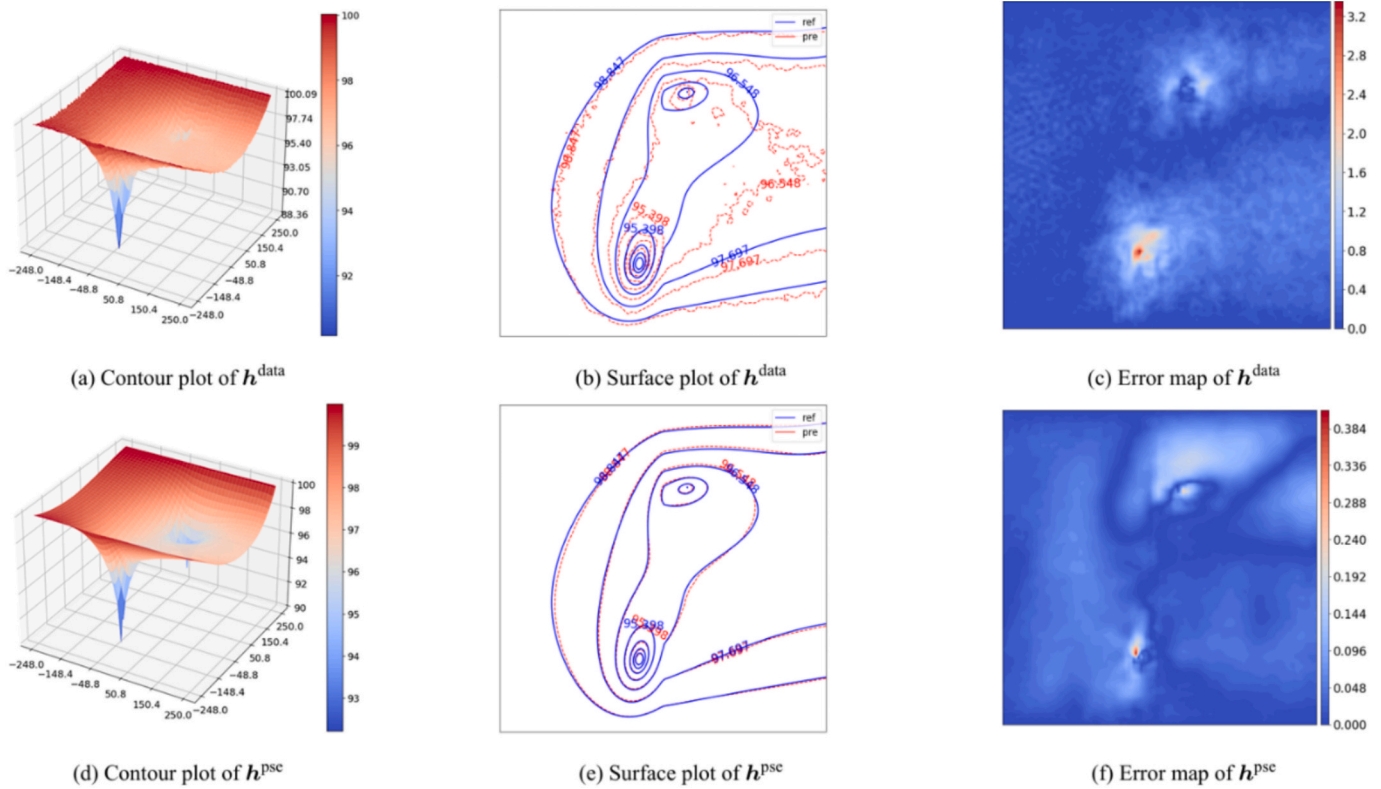


Fig. 12. The comparison of predicted results for Case 4 involving 2 pumping wells randomly distributed within the computational domain. Their pumping capacities are 785.0 and 696.6 m³/d, respectively.

model, we present a comparison between the predicted solutions obtained by the model trained on four distinct cases and the numerical solutions computed using the Finite Volume Method (FVM).

5.4.1. Cases 1–2: predictions with varying pumping capacity under fix well location

The predicted results both from the data-driven and the proposed methods with randomly pumping capacity, which is not shown in the training data, under fix location (as shown in Case 1 and Case 2) are shown in Figs. 9 and 10.

The surface plots (see Fig. 9(a), (d), Fig. 10(a), and (d)) and contour plots (see Fig. 9(b), (e), Fig. 10(b), and (e)) of both methods are nearly identical to the reference solutions. Notably, although our method does not utilize any true labeled data, its predicted results are visually highly similar to those of the data-driven method. While the absolute error of our method, as illustrated in the error plots for Case 1 (see Fig. 9(c) and (f)), is slightly higher than that of the data-driven method, its overall performance remains within an acceptable range. This demonstrates that our method can serve as an effective alternative to the data-driven approach. Furthermore, in Case 2 (see Fig. 10(c) and (f)), the error of our method is significantly lower than that of the data-driven method, highlighting its superior performance in more complex scenarios.

5.4.2. Cases 3–4: predictions with varying well numbers, locations, pumping capacities

We further validate the performance of our method in more challenging scenarios. The prediction results with randomly well locations and pumping capacities under fix 4 well numbers (Case 3) are shown in Fig. 11, and the prediction results with randomly well numbers, locations and pumping capacities (Case 4) are shown in Fig. 12. All the settings for these two scenarios are not shown in the training dataset. First, examining the surface plots, as shown in Fig. 11(d) and Fig. 12(d), the results generated by our method are almost identical to the reference

solutions for both scenarios, demonstrating excellent stability. In contrast, the surface plots produced by the data-driven method (see Fig. 11(a) and Fig. 12(a)) exhibit significant errors in high-gradient regions and near boundaries, with noticeable loss of detail and a marked decline in solution stability. Second, turning to the contour plots, Fig. 11(e) and Fig. 12(e) reveal that the contour plots generated by our method align almost perfectly with the reference solutions, indicating that our approach maintains very high accuracy even when handling complex cases. On the other hand, the contour plots from the data-driven method (see Fig. 11(b) and Fig. 12(b)) show significant deviations and noticeable noise, reflecting the method's instability and poor generalization capability. Finally, analyzing the error plots, Fig. 11(f) and Fig. 12(f) demonstrate that the error distributions of our method are smooth and exhibit low error levels across the entire solution domain, highlighting its robustness. In contrast, the error plots of the data-driven method (see Fig. 11(c) and Fig. 12(c)) display large and unacceptable errors, further emphasizing its limitations. Through the analysis of these two complex cases, we can conclude that our method exhibits stronger adaptability and higher accuracy when addressing high-complexity problems. It effectively mitigates the impact of noise and instability on the results, demonstrating significant performance advantages.

Furthermore, we present additional prediction results of our model (Case 4), including surface and contour plots that illustrate the impact of varying the number of wells from 3 to 8 and their different locations, as shown in Fig. 13. These visualizations clearly demonstrate the significant influence of well layout on the model's solutions, providing a critical basis for analyzing how the number and placement of wells affect simulation outcomes.

6. Conclusion and discussion

In this study, we propose a pseudo-label generator based on the traditional finite volume method (FVM) and the Jacobi iterative method

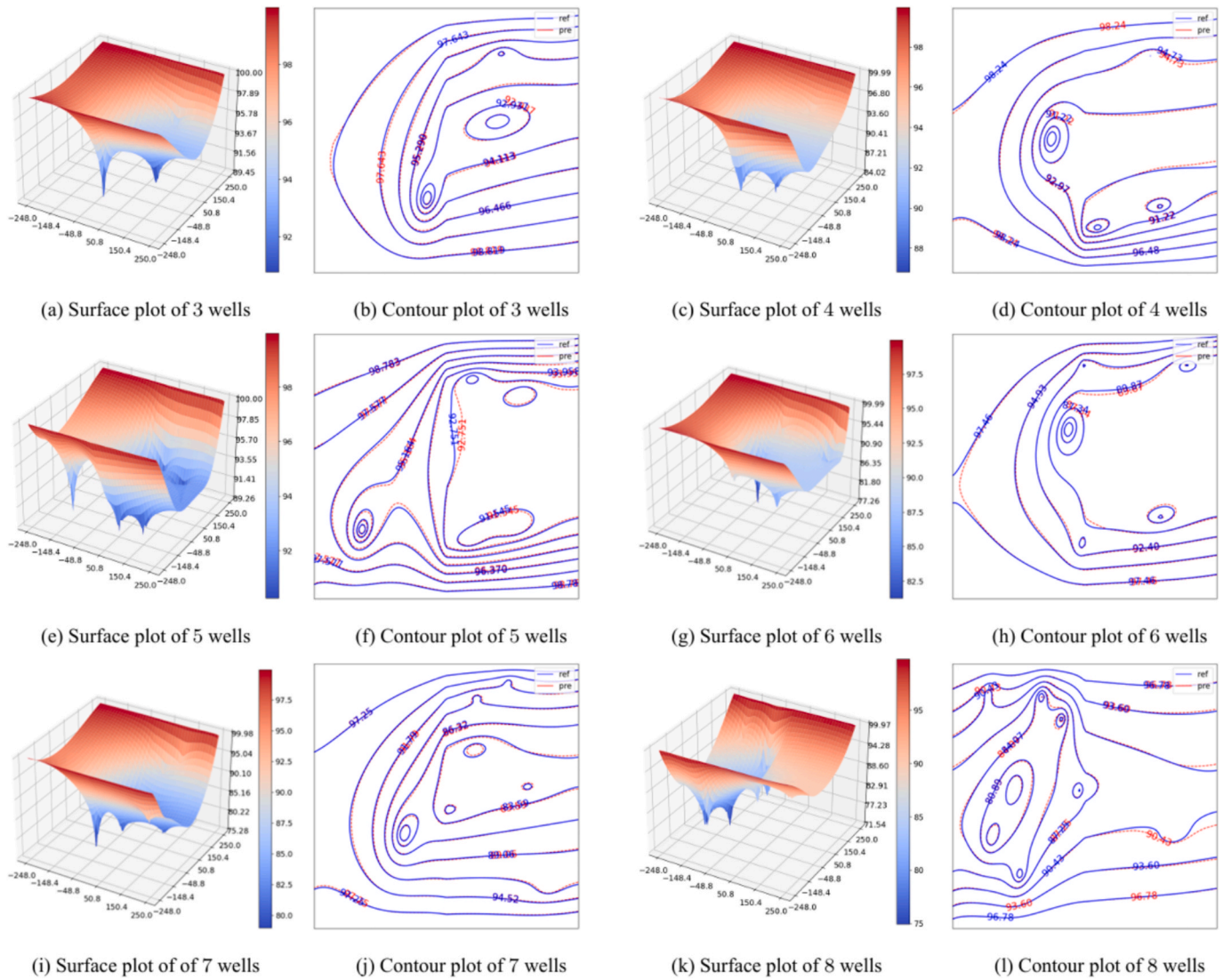


Fig. 13. The supplementary comparison of predicted results for Case 4, showcasing the predicted outcomes for scenarios involving 3 to 8 wells randomly distributed within the computational domain, including surface plots and error maps. For brevity, the pumping capacities are omitted here.

for the unsupervised training of a UNet. This approach leverages pseudo-labels derived from conventional numerical schemes, allowing the network to be trained without reliance on predefined labeled datasets. Numerical results demonstrate that our deep surrogate model performs comparably to data-driven models in simple scenarios and even surpasses them in more complex cases. Based on these results, we draw the following conclusions.

- (1) By incorporating physical constraints into the training process, our method significantly improves the accuracy than that of data-driven approaches in complex parameter spaces. This presents a powerful new paradigm for integrating deep learning with traditional numerical methods, effectively bridging the gap between data-driven and physics-driven modeling.
- (2) Extensive numerical experiments confirm the effectiveness of our method across a wide range of scenarios, highlighting its robustness and strong generalization capability.
- (3) The framework is highly flexible and can be readily extended to other numerical schemes and more complex parameterized equations, making it well-suited for a broad class of problems involving parameterized partial differential equations (PDEs). For solving the other parametrized problems, such as the

diffusion problem with parametrized K , new embedding methods should be developed for embedding the diffusion coefficient into images. Developing novel neural network architecture is also optional to take more complex parameters as input.

- (4) The proposed method is fully unsupervised, showcasing the potential of traditional numerical schemes to enhance the expressive power of neural networks. This provides a novel perspective for constructing surrogate models in future groundwater-related applications.

In future work, we plan to integrate this method with optimization problems to further expand its range of applications. Additionally, we aim to extend the approach to time-dependent problems to address a broader spectrum of challenges in scientific and engineering domains.

CRediT authorship contribution statement

Junqing Jia: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Yan Zhu:** Writing – review & editing. **Xiaoping Zhang:** Writing – review & editing, Methodology, Investigation. **Lili Ju:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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